Updated November 11, 2026

Homework problems for AMAT 540A (Topology I), Fall 2026. Over the course of the semester I'll add problems to this list, with each problem's due date specified. Each problem is worth 2 points.

Solutions will be gradually added (and may be hastily written without proofreading).

Problem 1 (due Weds 9/2): Let $f\colon A \to B$ be a function. Suppose that for all $A_0 \subseteq A$, $f^{-1}(f(A_0)) = A_0$. Prove that f is injective.

Solution: Let $a, a' \in A$ with $f(a) = f(a')$. Set $A_0 = \{a\}$, so $a' \in f^{-1}(f(A_0))$. By hypothesis $a' \in A_0$, so $a = a'$. □

Problem 2 (due Weds 9/2): Let A be a set and let $p_1\colon A \times A \to A$ be the projection function $p_1(a, a') = a$. Let $R \subseteq A \times A$ be a relation such that the image $p_1(R)$ equals all of A. Prove that if R is symmetric and transitive then it is reflexive.

Solution: Suppose R is symmetric and transitive. Let $a \in A$. Since $p_1(R) = A$ there exists $(a_1, a_2) \in R$ with $p(a_1, a_2) = a$. By construction of p this means $a_1 = a$, so a is related to a_2 under R. By symmetry a_2 is related to a, and by transitivity a is related to a, so we conclude R is reflexive. □

Problem 3 (due Weds 9/2): Let $\mathcal{A}$ be a collection of sets and let $U = \bigcup_{A \in \mathcal{A}} A$. Prove that if $\mathcal{P}(U) = \bigcup_{A \in \mathcal{A}} \mathcal{P}(A)$ then $U \in \mathcal{A}$.

Solution: Since $U \in \mathcal{P}(U)$, our hypothesis ensures that $U \in \bigcup_{A \in \mathcal{A}} \mathcal{P}(A)$, i.e., $U \in \mathcal{P}(A)$ for some $A \in \mathcal{A}$. This means $U \subseteq A$, and $A \subseteq U$ by definition, so we conclude $U = A \in \mathcal{A}$ as desired. □

No class Monday 9/7 (Labor Day)

Problem 4 (due Weds 9/9): Prove that $\{U \subseteq \mathbb{N} \mid U = \emptyset \text{ or } |\mathbb{N} \setminus U| < 2026\}$ is not a topology on $\mathbb{N}$.

Solution: Let $U = \mathbb{N} \setminus \{1, \ldots, 2025\}$ and $V = \mathbb{N} \setminus \{2026\}$, so U, V are both in this set, but $U \cap V = \mathbb{N} \setminus \{1, \ldots, 2026\}$ is not in this set. □

Problem 5 (due Weds 9/9): Let X be a set and $\sim$ an equivalence relation. Write $[x]$ for the equivalence class of $x \in X$ with respect to $\sim$. Prove that $\mathcal{B} = \{[x] \mid x \in X\}$ is a basis for a topology on X.

Solution: Since $x \in [x]$ for all x we have the first axiom. Now suppose $x \in [y] \cap [z]$. Then $x \sim y$ and $x \sim z$, so $[y] \cap [z] = [x] \cap [x] = [x]$, so $x \in [x] \subseteq [y] \cap [z]$ reveals the second axiom. $\square$

Problem 6 (due Weds 9/9): Let $\mathcal{T} = \{U \subseteq \mathbb{R} \mid U = \emptyset \text{ or } \mathbb{R} \setminus U \text{ is countable}\}$. Is $\mathcal{T}$ a topology on $\mathbb{R}$? Prove or disprove.

Solution: Yes. First note that $\emptyset \in \mathcal{T}$ by construction, and $\mathbb{R} \in \mathcal{T}$ since $\mathbb{R} \setminus \mathbb{R} = \emptyset$ is countable. Next for any family $U_\alpha \in \mathcal{T}$, without loss of generality none empty, we have that $\mathbb{R} \setminus U_\alpha$ is countable for all α, so $\mathbb{R} \setminus (\bigcup_\alpha U_\alpha)$ is countable (being a subset of any of these), hence $\bigcup_\alpha U_\alpha \in \mathcal{T}$. Finally, for $U, V \in \mathcal{T}$, if one of them is empty their intersection is empty hence in $\mathcal{T}$, and otherwise $\mathbb{R} \setminus U$ and $\mathbb{R} \setminus V$ are countable, hence $\mathbb{R} \setminus (U \cap V) = (\mathbb{R} \setminus U) \cup (\mathbb{R} \setminus V)$ is countable, so $U \cap V \in \mathcal{T}$. $\square$

Problem 7 (due Weds 9/16): Let X be an infinite set with the finite complement topology. Prove that $\mathcal{B} := \{B \subseteq X \mid |X \setminus B| \text{ is a prime number}\}$ is a basis for this topology.

Solution: Let U be open in X and $x \in U$. Since $U \neq \emptyset$, $|X \setminus U| < \infty$. Choose a prime number p greater than $|X \setminus U|$. Since X is infinite so is $U \setminus \{x\}$, hence we can choose $P \subseteq U \setminus \{x\}$ such that $|X \setminus U| + |P| = p$. Now setting $B := U \setminus P$ we get $|X \setminus B| = |X \setminus U| + |P| = p$, so $B \in \mathcal{B}$, and $x \in B \subseteq U$, so $\mathcal{B}$ is a basis. $\square$

Problem 8 (due Weds 9/16): Let X be a set with at least two elements, with a total order $\leq$ such that for all $a < b$ in X there exists $c \in X$ with $a < c < b$. (Here $<$ means "$\leq$ and $\neq$".) Prove that the order topology on X is not the same as the discrete topology.

Solution: Since X has at least two elements, the other hypothesis ensures that there exists $x \in X$ that is neither a global min nor max. We claim $\{x\}$ is not open in the order topology, hence the order topology is not discrete. Suppose $\{x\}$ is open, so either $\{x\} = (a, b)$ for some $a < b$, or $\{x\} = [a_0, b)$ for some global min a_0 and some b, or $\{x\} = (a, b_0]$ for some a and some global max b_0. Since $a_0 \in [a_0, b)$ and $b_0 \in (a, b_0]$, but x is not a global min or max, these cases cannot happen, so $x = (a, b)$ for some $a < b$. Hence $a < x < b$, but the hypotheses ensure there exist $c, d \in X$ with $a < c < x$ and $x < d < b$, so (a, b) has more elements than just x in it, a contradiction. $\square$

Problem 9 (due Weds 9/16): Let X be the set from Problem 8, with total order $\leq_X$ having the same hypotheses (so the order topology is not discrete). Let Y be another totally ordered set with order $\leq_Y$ and at least two elements. Let $\leq_{lex}$ be the lexicographic order on $X \times Y$.

Prove that the order topology on $X \times Y$ using $\leq_{lex}$ does not equal the product topology using $\leq_X$ and $\leq_Y$.

Solution: [Let me write $\langle x, y\rangle$ for ordered pairs, to distinguish them from open intervals (a, b).] Since X is as in Problem 8, we know we can choose some $a < x < b$ in X. Let $y_1 < y_2$ in Y. Consider the lexicographic interval $L = (\langle x, y_1\rangle, \langle b, y_2\rangle)$, so L is open in the order topology. Note that L contains the point $\langle x, y_2\rangle$. Now we claim that no basic open set $U \times V$ in the product topology can satisfy $\langle x, y_2\rangle \in U \times V \subseteq L$. Indeed, this would mean that $x \in U$ and hence we could choose an interval (x_1, x_2) in X with $x \in (x_1, x_2) \subseteq U$. Now choose x_3 with $x_1 < x_3 < x$, so the point $\langle x_3, y_2\rangle$ lies in $U \times V$, but $\langle x_3, y_2\rangle$ is not lexicographically greater than $\langle x, y_1\rangle$, contradicting that $U \times V \subseteq L$. □

Exam in class on Weds 9/23

Problem 10 (due Weds 9/23): Let X and Y be topological spaces and consider $X \times Y$ with the product topology. Prove that if $C \subseteq X$ and $D \subseteq Y$ are closed then $C \times D$ is closed.

Solution: Let $C \subseteq X$ and $D \subseteq Y$ be closed. Note that $(X \times Y) \setminus (C \times D) = ((X \setminus C) \times Y) \cup (X \times (Y \setminus D))$. Since C and D are closed, $X \setminus C$ and $Y \setminus D$ are open, so $(X \setminus C) \times Y$ and $X \times (Y \setminus D)$ are open in the product topology, hence so is their union. We conclude that $C \times D$ is closed. □

Problem 11 (due Weds 9/23): Let X and Y be infinite sets with the finite complement topology. Prove that the finite complement topology on $X \times Y$ is not a refinement of the product topology on $X \times Y$.

Solution: We must exhibit a subset that is open in the product topology but not open in the finite complement topology. Equivalently, taking complements, we must exhibit a subset that is closed in the product topology but not closed in the finite complement topology. Fix $y_0 \in Y$ and consider $X \times \{y_0\}$. Since Y has the finite complement topology, $\{y_0\}$ is closed, and so by the previous problem $X \times \{y_0\}$ is closed in the product topology. Since X is infinite $X \times \{y_0\}$ is infinite, and since Y is infinite $X \times \{y_0\}$ is not all of $X \times Y$. Thus $X \times \{y_0\}$ is not closed in the finite complement topology. □

Problem 12 (due Weds 9/23):
Let X be a set with at least two elements, with the *excluded point topology* $\{U \mid \spadesuit \notin U\} \cup \{X\}$, where $\spadesuit$ is some fixed element. (It's pretty obvious this is a topology, you don't need to bother proving that.) Prove that X has exactly two sets that are simultaneously open and dense.

Solution: Of course X itself is open and dense. Also, $X \setminus \{\spadesuit\}$ is open by definition, and is non-closed (again by definition), so its closure is X since this is the only closed subset containing it. This shows that X and $X \setminus \{\spadesuit\}$ are open and dense, and now we must show

no other open subsets are dense. Let $U \neq X$ be open, so $\spadesuit \notin U$. Note that $U \cup \{\spadesuit\}$ is closed, and U is not, so $\overline{U} = U \cup \{\spadesuit\}$. Now the only way U can be dense is if $U \cup \{\spadesuit\} = X$, i.e., if $U = X \setminus \{\spadesuit\}$. $\square$

Problem 13 (due Weds 9/30): Let X be a topological space and $S \subseteq Y \subseteq X$. Suppose that S is dense in Y using the subspace topology on Y, and Y is dense in X. Prove that S is dense in X.

Solution: Let U be a non-empty open set in X. Since Y is dense in X we know $Y \cap U \neq \emptyset$. Since U is open in X we know $Y \cap U$ is open in Y. Since S is dense in Y we know $(Y \cap U) \cap S \neq \emptyset$. In particular $U \cap S \neq \emptyset$. We conclude that S is dense in X. $\square$

Problem 14 (due Weds 9/30): Let X be the topological space from Problem #5. Prove that if X is Hausdorff then it is discrete.

Solution: Suppose X is Hausdorff. Let $x \neq y$ in X. Since X is Hausdorff we can separate x and y with disjoint basic open neighborhoods, say $x \in B_x$ and $y \in B_y$ for basic open set B_x and B_y with $B_x \cap B_y = \emptyset$. By the definition of the basis, $B_x = [x]$ and $B_y = [y]$, so $B_x \cap B_y = \emptyset$ implies $x \not\sim y$. This worked for all $x \neq y$, so $\sim$ is just the equivalence relation $=$, and each basic open set is a singleton, proving that the topology is discrete. $\square$

Problem 15 (due Weds 9/30): Let X and Y be topological spaces and let $f\colon X \to Y$ be an injective function. Suppose X is Hausdorff and Y has the finite complement topology. Prove that f is continuous.

Solution: Let $C \subseteq Y$ be closed. If $C = Y$ then $f^{-1}(C) = X$ is closed. If $C \neq Y$ then C is finite, and since f is injective $f^{-1}(C)$ is also finite. Since X is Hausdorff, $f^{-1}(C)$ is closed. Having shown that the preimage of every closed set is closed, we conclude that f is continuous. $\square$

Problem 16 (due Weds 10/7): Let X and Y be topological spaces and let $y_0 \in Y$. Let $\iota\colon X \to X \times Y$ be $\iota(x) := (x, y_0)$, and view $X \times Y$ with the product topology. Prove that ι is continuous.

Solution: Let $U \times V$ be a basic open set in $X \times Y$, so U is open in X and V is open in Y. If $y_0 \in V$ then $\iota^{-1}(U \times V) = U$, which is open, and if $y_0 \notin V$ then $\iota^{-1}(U \times V) = \emptyset$, which is open. $\square$

Problem 17 (due Weds 10/7): Let Λ be an infinite set. Let $(X_\alpha)_{\alpha \in \Lambda}$ be a family of topological spaces and let $X = \prod_{\alpha \in \Lambda} X_\alpha$ be their product, with the product topology. For each $\alpha \in \Lambda$ let $U_\alpha \subseteq X_\alpha$ be open, and suppose that $\emptyset \neq U_\alpha \neq X_\alpha$ holds for all $\alpha \in \Lambda$. From the definition

of product topology, we know that $\prod_{\alpha\in\Lambda} U_\alpha$ is not a basic open set. Prove that it is not even an open set at all.

Solution: Since $\emptyset \neq U_\alpha$ holds for all α we know $\prod_{\alpha\in\Lambda} U_\alpha$ is non-empty. Thus to prove it is not open, it suffices to prove it contains no basic open sets. For a basic open set $\prod_{\alpha\in\Lambda} V_\alpha$ we know that each V_α is open in X_α, and for all but finitely many α we have $V_\alpha = X_\alpha$. Since Λ is infinite, we can choose α with $V_\alpha = X_\alpha$. Since $U_\alpha \neq X_\alpha$ we know $V_\alpha \not\subseteq U_\alpha$, and hence $\prod_{\alpha\in\Lambda} V_\alpha \not\subseteq \prod_{\alpha\in\Lambda} U_\alpha$. □

Problem 18 (due Weds 10/7): Let $C = \prod_{\mathbb{N}} \{0,1\} = \{0,1\} \times \{0,1\} \times \cdots$ be the Cantor space. Let C_0 be the basic open set $C_0 := \{0\} \times \{0,1\} \times \{0,1\} \times \cdots$. Prove that $C_0 \cong C$. [For fun, you can try to prove that every basic open set is homeomorphic to C (this might have an annoying number of cases though, I'm not sure).]

Solution: Let $f\colon C_0 \to C$ send $(0, a_2, a_3, \dots)$ to $(a_2, a_3, \dots)$, and let $g\colon C \to C_0$ send $(b_1, b_2, \dots)$ to $(0, b_1, b_2, \dots)$. Clearly $f \circ g$ and $g \circ f$ are the identity functions, so f is a bijection. For any basic open set $U_1 \times U_2 \times \cdots$ in C, the preimage under f is $\{0\} \times U_1 \times U_2 \times \cdots$, which is open in C and hence open in C_0 (since C_0 is open in C). Thus f is continuous. Similar, for any basic open set $\{0\} \times U_1 \times U_2 \times \cdots$ in C_0 (indeed these are the basic open sets in C_0, being exactly the intersections of C_0 with the basic open sets in C), the preimage under g is $U_1 \times U_2 \times \cdots$, which is open in C. Thus $g = f^{-1}$ is continuous, and we are done. [To prove this for an arbitrary basic open set, I think induction makes it not too bad: induct on the largest i such that $U_i \neq \{0,1\}$, so the base case holds trivially, and for the induction step do a "shift" map like above.] □

No class Monday 10/12 (October Break)

Problem 19 (due Weds 10/14): Let $p\colon X \to Y$ be a quotient map. Prove that if X is discrete then so is Y.

Solution: Let $A \subseteq Y$ be arbitrary. Since X is discrete, $p^{-1}(A)$ is open in X, and since p is a quotient map this implies A is open in Y. □

Problem 20 (due Weds 10/14): Let X be a topological space and $U \subseteq X$ an open subset. Let X/U be the quotient space where we crush U to a point. Prove that if X/U is metrizable then U is closed.

Solution: Let $p\colon X \to Y := X/U$ be the quotient map and let $y_U := p(U)$. Since Y is metrizable it is Hausdorff, so $\{y_U\}$ is closed. Since p is continuous, $U = p^{-1}(y_U)$ is also closed. □

Problem 21 (due Weds 10/14): Let (X, d) be a metric space. Fix a point $x_0 \in X$ and let $f\colon X \setminus \{x_0\} \to (0, \infty)$ be the function $f(x) = d(x, x_0)$. Prove that f is continuous (using the subspace topology on $X \setminus \{x_0\}$ and the usual topology on $(0, \infty)$). [Just a warning: We never talked about "closed" balls in X, and we don't officially know yet that they are closed as subsets of X, so if you use a closed ball at some point make sure to prove it's closed. [Additional warning: The closure of an open ball doesn't necessarily equal the analogous closed ball!]]

Solution: We must show that the preimage of an arbitrary basic open set (a, b) is open $(0 \leq a < b)$. This preimage equals $\{x \in X \mid a < d(x, x_0) < b\}$. This equals the open ball $B_b(x_0)$ with the "closed ball" $C_a(x_0) := \{y \in X \mid d(y, x_0) \leq a\}$ removed, so it suffices to prove that $C_a(x_0)$ is closed. For $y \in X \setminus C_a(x_0)$ let $r := d(y, x_0) - a$, so $r > 0$, and we claim that $B_r(y) \subseteq X \setminus C_a(x_0)$. Let $z \in B_r(y)$, so $d(z, y) < r$, i.e., $d(z, y) < d(y, x_0) - a$. Now $d(z, x_0) \geq d(y, x_0) - d(z, y) > a$, so $z \in X \setminus C_a(x_0)$. We conclude that $X \setminus C_a(x_0)$ is open, so $C_a(x_0)$ is closed. $\square$

No class Weds 10/21 (I'm out of town) and no homework due that day

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Exam in class on Weds 10/28.

Problem 22 (due Weds 10/28): Prove that every topological space is homeomorphic to a subspace of a connected topological space.

Solution: Let X be a topological space with topology $\mathcal{T}$. Set $Y = X \cup \{y_0\}$ for some new point $y_0$and set $\mathcal{T}_Y = \mathcal{T} \cup \{Y\}$. This is a topology on Y, and the subspace topology on X equals $\mathcal{T}$, so we just need to prove Y is connected. If $U \subseteq Y$ then either $y_0 \in U$ or $y_0 \in Y \setminus U$. Thus if these are both open in Y, one of them must equal all of Y. $\square$

Problem 23 (due Weds 10/28): Let $X = \mathbb{Z}$ with the finite complement topology. Prove that X is connected but not path connected. [The first thing should be easy. For the second thing, feel free to use the known fact that if $\{C_\alpha\}_{\alpha \in \Lambda}$ is a collection of pairwise disjoint proper non-empty closed subsets of $[0, 1]$ whose union is all of $[0, 1]$, then Λ is uncountable.] [Looks like I forgot to write "proper", but people figured it out.]

Solution: A proper non-empty clopen set would have to be finite and have finite complement, which is impossible since $\mathbb{Z}$ is infinite. Now let $x, y \in X$, and suppose there exists a map $p\colon [0, 1] \to X$ with $p(0) = x$ and $p(1) = y$. For each x in the image of p, let $C_x = p^{-1}(\{x\})$, so the C_x are pairwise disjoint non-empty subsets of $[0, 1]$ whose union is $[0, 1]$. Also, each $\{x\}$ is closed in X, so by continuity C_x is closed. By the "known fact", since X is countable, there can only be one C_x and it equals all of $[0, 1]$, i.e., p is constant, so $x = y$. Thus the path components are singletons, and X is not path connected. $\square$

Problem 24 (due Weds 10/28): Let $(X_\alpha)_{\alpha\in\Lambda}$ be a family of path connected spaces. Let $f, g \in \prod_{\alpha\in\Lambda} X_\alpha$, and for each α let $p_\alpha\colon [0,1] \to X_\alpha$ be a (continuous) path from $f(\alpha)$ to $g(\alpha)$. Let $p\colon [0,1] \to \prod_{\alpha\in\Lambda} X_\alpha$ be the function sending t to the function $p(t)$ defined by $p(t)(\alpha) := p_\alpha(t)$ for all α. Prove that p is continuous. (So this proves that $\prod_{\alpha\in\Lambda} X_\alpha$ is path connected.)

Solution: We actually already basically did this in class, I won't slog through the proof. □

Problem 25 (due Weds 11/4): Prove that $\mathbb{R}$ with the finite complement topology is path connected.

Solution: Let $a, b \in \mathbb{R}$. Let $p\colon [a,b] \to \mathbb{R}$ be $p(x) = x$. Since $[a,b]$ is Hausdorff, Problem #15 says p is continuous. □

Problem 26 (due Weds 11/4): Let X be a locally connected topological space. Prove that every open subspace of X is locally connected.

Solution: Let U be open in X and let $x \in U$. Since X is locally connected, we can choose open V in X with $x \in V \subseteq U$. Since $V = V \cap U$, V is also open in U, and so we are done. □

Problem 27 (due Weds 11/4): Let X be a locally path connected topological space. Prove that every connected component is locally connected.

Solution: Let C be a connected component of X. Let $x \in C$. Since X is locally path connected, we can choose a path connected, hence connected, open $V \subseteq X$ with $x \in V$. Since $V \cap C \neq \emptyset$ and V is connected, $V \subseteq C$. Now $V = V \cap C$ shows V is open in C, and so we conclude C is locally connected. □

Problem 28 (due Weds 11/11): Let X be a Hausdorff space. Let $\{K_\alpha\}_{\alpha\in\Lambda}$ be a family of compact subspaces of X. Prove that $\bigcap_{\alpha\in\Lambda} K_\alpha$ is compact. [I don't think I already proved this in class, but if I did, free points.]

Problem 29 (due Weds 11/11): Let P be the set of all prime numbers. For $S \subseteq \mathbb{N}$ let $P(S) \subseteq P$ be the set of all prime divisors of elements of S. Let $X = \mathbb{N}$ with the topology $\mathcal{T} = \{U \subseteq X \mid |P(X \setminus U)| < \infty\} \cup \{\emptyset\}$. Prove that X is not compact.

Problem 30 (due Weds 11/11): Let X be a locally compact Hausdorff space. Let $Y = X \cup \{\infty\}$ with the topology $\{U \subseteq X \mid U \text{ is open in } X\} \cup \{Y \setminus C \mid C \subseteq X \text{ is closed in } X\}$. (So

this uses "closed" instead of "compact" for C. And trust me that this is a topology.) Prove that if Y is compact, then X was already compact. (So this is why this is the "wrong" way to try and define a one-point compactification.)

Problem 31 (due Weds 11/18): Let X be a second countable Hausdorff space. Suppose X admits a basis in which every basic open set is compact. Prove that X is metrizable.

Problem 32 (due Weds 11/18): Let $X = \mathbb{Z}$ with the particular point topology, i.e., a non-empty subset is open iff it contains 0. Let $c\colon X \to X$ be the constant map $c(x) = 0$. Prove that every map $f\colon X \to X$ is homotopic to c.

Problem 33 (due Weds 11/18): Let $C = \prod_{\mathbb{N}} \{0, 1\}$ be the usual Cantor space (see for example Problem #18). Given a topological space X, call a continuous map $f\colon C \to X$ a *Cantor path* in X. Given Cantor paths $f, g\colon C \to X$, define $(f * g)\colon C \to X$ via $(f * g)(\varepsilon_1, \varepsilon_2, \varepsilon_3, \dots) = f(\varepsilon_2, \varepsilon_3, \dots)$ if $\varepsilon_1 = 0$ and $(f * g)(\varepsilon_1, \varepsilon_2, \varepsilon_3, \dots) = g(\varepsilon_2, \varepsilon_3, \dots)$ if $\varepsilon_1 = 1$. Prove that $(f * g)$ is continuous, hence a Cantor path. [I kind of just made this up, I don't know if it could lead to some notion of "Cantor groupoid", or at least "Cantor category" (forgetting invertibility), but, maybe?]

No class Weds 11/25 (Thanksgiving Eve)

Exam in class on Weds 12/2.

Problem 34 (due Weds 12/2):

Problem 35 (due Weds 12/2):

Problem 36 (due Weds 12/2):
